

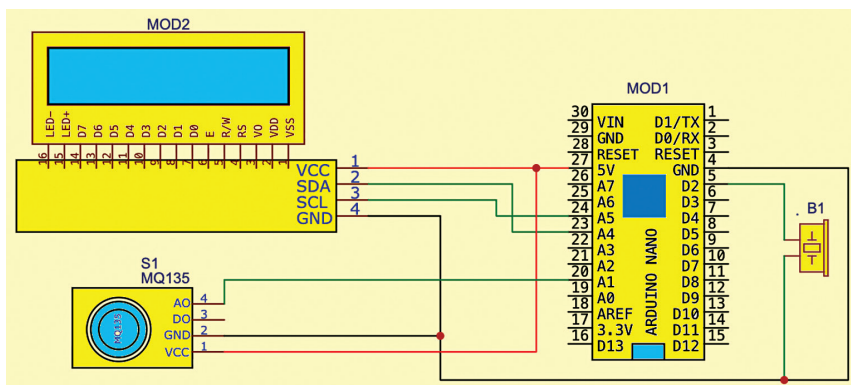


Fig. 2: Circuit diagram

```

sketch_dec24a.ino
1  #include <LiquidCrystal_I2C.h>
2  LiquidCrystal_I2C lcd(0x27,16,2);
3  #define sensor A1
4  void setup(){
5    lcd.init();
6    lcd.clear();
7    lcd.backlight(); // Make sure backlight is on
8    lcd.setCursor(0,0);
9    lcd.print("Air Quality");
10   lcd.setCursor(0,1);
11   lcd.print("Monitoring");
12   pinMode(sensor,INPUT);
13   pinMode(2,OUTPUT);
14
15   delay(2000);
16   lcd.clear();

```

Fig. 3: Code snippet showing I²C address and LCD configuration

predefined threshold values. When air quality exceeds the safe limit, indicating hazardous conditions, the buzzer activates. Under normal operating conditions, the buzzer remains off.

Software

To interface the I²C LCD, the required library must be installed using the Arduino Library Manager. Search for LiquidCrystal_I2C.h and install the appropriate display library. Within the program, the LCD type is configured, and its I²C address is specified. The address can be obtained from the display vendor or the datasheet; commonly used I²C LCD modules operate at address 0x27.

If the display does not function correctly or the address is unknown, the I²C (wire) scanner sketch available in the Arduino IDE examples can be used to identify the correct

address. Within the main program loop, the system continuously reads the MQ135 sensor output and updates the air quality values displayed on the LCD. An 'if' statement compares the sensor reading against predefined thresholds; when toxic air levels are detected, the buzzer activates to alert the user.

Construction and testing

The source code is first uploaded to the Arduino Nano, after which the system is assembled on a breadboard. Fig. 1 should be consulted before assembly to understand component placement and the overall layout. All connections must be made strictly according to the circuit diagram shown in Fig. 2 to ensure reliable operation.

The MQ135 gas sensor must be positioned directly in the ambient

air to ensure accurate sensing. Once system operation is verified on the breadboard, the assembly can be transferred to a general-purpose PCB for permanent installation. The Arduino Nano may be mounted using female header pins to allow easy replacement or reprogramming, if required. The MQ135 sensor should be mounted near the edge of the PCB or aligned with the enclosure's ventilation slots to ensure proper air circulation.

A ventilated plastic enclosure is recommended to protect the electronics while maintaining unobstructed airflow to the sensor.

For testing, the system is powered using a USB cable connected to a computer or a regulated 5V power supply. After power-up, the MQ135 sensor should be allowed a few minutes to warm up and stabilise. The LCD continuously displays the air quality status. Introducing smoke or alcohol vapour near the sensor results in a noticeable change in the displayed readings. When the air quality reaches the predefined very poor or toxic range, the buzzer activates, confirming correct system operation.

This compact, low-cost air quality monitoring system is well-suited for indoor air monitoring, educational demonstrations, and DIY applications.

For outdoor use, connect a 9V battery instead of a USB supply, making it portable and independent. The battery should be connected to the Vin pin of the Arduino Nano. This arrangement is suitable for field and temporary installations where USB power is not available and can reliably run the controller, sensors, and display. **EFY**

TO DOWNLOAD SOURCE CODE AND RESOURCES



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